

Project Specification & UI Architecture Report

A production-grade, desktop-first responsive travel planning web application featuring real-time budget balancing, verified local services, and multimodal transport intelligence.

1. Executive Overview & Problem Statement

Modern Indian travellers face highly fragmented platforms when discovering local destinations. Standard booking portals push sponsored listings, conceal public transportation options, and lack integrated itinerary budget calculators. **TRAAR** is built for the Smart India Hackathon 2026 to introduce transparent, verified tariffs across four foundational pillars: **Stays**, **Food**, **Places to Visit**, and **Nearby Trips**, powered by a persistent trip balancing engine.

2. Design System & Category Accent Matrix

Category / Element	Accent Color	Hex Code	Applied Context & UI Semantics
Sidebar Shell	Deep Navy Teal	#0A2624	Fixed 235px navigation with SIH pilot badge and active filled pill
Stays	Amber Gold	#D97706	Hotels, hostels, PGs, dormitories, tariff units, and way-to modals
Food & Dining	Vibrant Orange	#EA580C	Bhopali delicacies, restaurants, cafes, dhabas, veg/non-veg chips
Places to Visit	Royal / Sky Blue	#0284C7	Heritage landmarks, lake vistas, entry tariffs, geo-map view
Nearby Trips	Royal Purple	#7C3AED	Regional excursions, distance badges, multimodal transport panel
Budget Warnings	Crimson Red	#DC2626	Warning banner: <i>'Tiny pause, you have reached your limit.'</i>

3. Complete 10-Page Architecture Breakdown

Page Name	Core Features & Functional Components	Key User Actions
1. Home Page	Hero with Bhopal Upper Lake sunset, destination dropdown selector, pilot notice, 4 category explore cards, popular cities.	Select State/City, Explore Destination, jump to categories.
2. Destination Overview	Upper Lake panorama banner, history & culture summary, quick info cards (Best time, Language, Famous for, How to reach), video tour modal.	Watch video guide, explore category cards, review pilot data notes.
3. Stays	3-column card grid of verified stays (Jehan Numa, Backpackers, PGs, Dorms), filter chips (Hotel, PG, Hostel), price & distance sliders, desktop My Trip summary panel.	View Details modal, Add to Trip, Way to Hotel modal, live subtotal.

4. Food	Bhopali culinary directory (Manohar Dairy, ICH, Sharma Dhaba, Hakeem), Pure Veg / Non-Veg segmented toggle, average cost per person, desktop summary drawer.	Filter cuisines, Add to Trip, Way to Restaurant transit directions.
5. Places to Visit	Segmented filters (Nearest First, Famous, Hidden Gems), list / geo-map view switch, attractions with verified entry tariffs and timings (Free to Rs. 50).	Interactive map pins, inspect operating hours, Add to Trip.
6. Nearby Trips	Distance filters (within 50 km, 50-100 km, 200+ km), excursion cards (Sanchi, Bhimbetka, Bhojpur, Ujjain, Pachmarhi), interactive right transport panel (Bus, Train, Car).	Compare operator tariffs, first/last departure, Add transport to trip.
7. My Trip Planner	Parameters bar (dates, travellers, entered budget), itemized rows with steppers (+/-), line subtotals, over-budget warning banner, review checkbox, Export Trip Estimate.	Balance budget, export PDF estimate, share trip link, adjust quantities.
8. Provider Dashboard	Status KPI metrics (All, Approved, Pending Review, Rejected, Draft, Withdrawn), listings management table with status pills, search & category filters.	Create New Listing CTA, preview listing, withdraw/delete service.
9. Add Service Wizard	4-step form (Basic Info, Pricing & Units, Additional Info, Photos & Verification), right-side progress tracker (0-100%), SIH validation tips.	Save as Draft, Preview Modal, Submit for Review (live updates dashboard).
10. Help Booth	Instant search bar, 9 FAQ accordions covering all requirements (selection, filters, freshness, budget, privacy), official SIH ticket support modal.	Search help articles, expand FAQ answers, submit support query.

4. Production UI Screen Designs (SIH 2026 Reference)

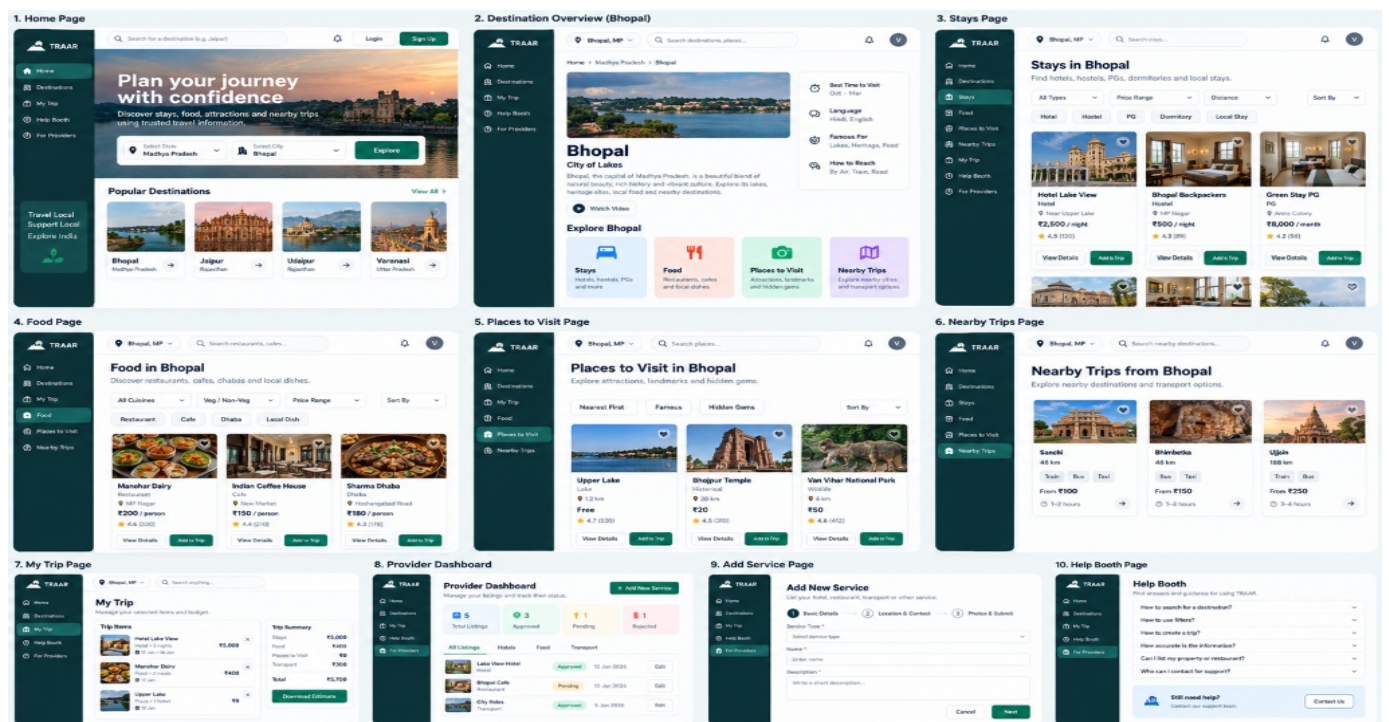


Figure 1: Full 10-screen UI layout rendered across Home, Overview, Stays, Food, Places, Nearby, My Trip, Provider Dashboard, Add Service, and Help Booth.

5. Dynamic Budget Balancing Engine & Alert Logic

The application maintains real-time mathematical consistency across all itinerary modifications. The grand total is dynamically computed as:

Grand Total = Total Stays + Total Food + Total Attractions + Total Transport

Where:

- **Total Stays** = Σ (Room / Bed Tariff $\times$ Duration $\times$ Quantity)
- **Total Food** = Σ (Average Spend per Person $\times$ Travellers $\times$ Meals)
- **Total Attractions** = Σ (Verified Entry Fee $\times$ Number of Visitors)
- **Total Transport** = Σ (Fare per Passenger / Vehicle $\times$ Booked Slots)

Over-Budget Alert Condition:

Whenever Grand Total > Entered Target Budget, an inline red warning banner renders reactively:

■■ Tiny pause, you have reached your limit.

Displaying the exact variance deficit without triggering disruptive browser modal dialogs.

6. Engineering Verification & Build Audit

Verification Check	Target Metric	Observed Result	Status
Vite Production Build	Zero errors / clean bundle	1,898 modules transformed in 766ms	PASSED
Development Server	Port 5173 host bound	Ready in 62ms at http://localhost:5173/	ONLINE
In-Browser PDF Generator	jsPDF + html2canvas	Generates downloadable A4 estimate with disclaimer	PASSED
Responsive Layout	Desktop, Tablet, Mobile	Fixed 235px sidebar with mobile drawer conversion	PASSED